Semiconductors

Technology / Research snapshot: 20 Jun 2025

Semiconductors are engineered materials and devices that control electrical current; the industry spans design, fabrication, equipment, packaging, and distribution.

What is happening?

Leading-edge logic demand is being pulled forward by AI accelerators, while memory is recovering from a deep inventory correction. The supply chain is becoming more regional, expensive, and policy-dependent.

Why it matters

Chips are the physical substrate for computing, communications, and industrial automation. A bottleneck in one layer can set the capacity and pricing power of the entire digital economy.

Risk assessment

The risk map includes Taiwan concentration, export controls, cyclical overbuild, and a widening gap between leading-edge demand and mature-node utilization.

What to monitor

01 Wafer-fab equipment bookings

02 HBM and DRAM contract pricing

03 Foundry utilization by node

04 Taiwan Strait risk indicators

Timeline

2020 Pandemic demand exposes inventory fragility

2022 Memory cycle turns down

2023 AI accelerators create a new scarcity

2025 Regional capacity race moves into construction

Sources and limitations

Sources listed in the original topic: WSTS, 2025 forecast, Company filings, NEXUS supply-chain model.

This edition reproduces the existing NexusNews topic research as of the date above. It is a structured synthesis. Figures on the topic page are illustrative.